

§2.4

Models and Strategies for Multiplication and Division

Many ways to multiply and divide, and the model each one comes from

Today's Goal

After completing this section, you should be able to:

- Use the equal groups, array, and area models to develop strategies for multiplying and dividing whole numbers
- Apply multiple strategies, including the partial products, standard, partial quotients, and long division algorithms, to multiply and divide whole numbers using appropriate representations

LT.6 Multiplication and Division Strategies for Whole Numbers

I can apply multiple strategies (including the standard algorithms) for multiplication and division of whole numbers using appropriate representations.

How Many Ways?

Take 2 minutes to think to yourself.

How many ways can you think of to efficiently evaluate 8×24 ?

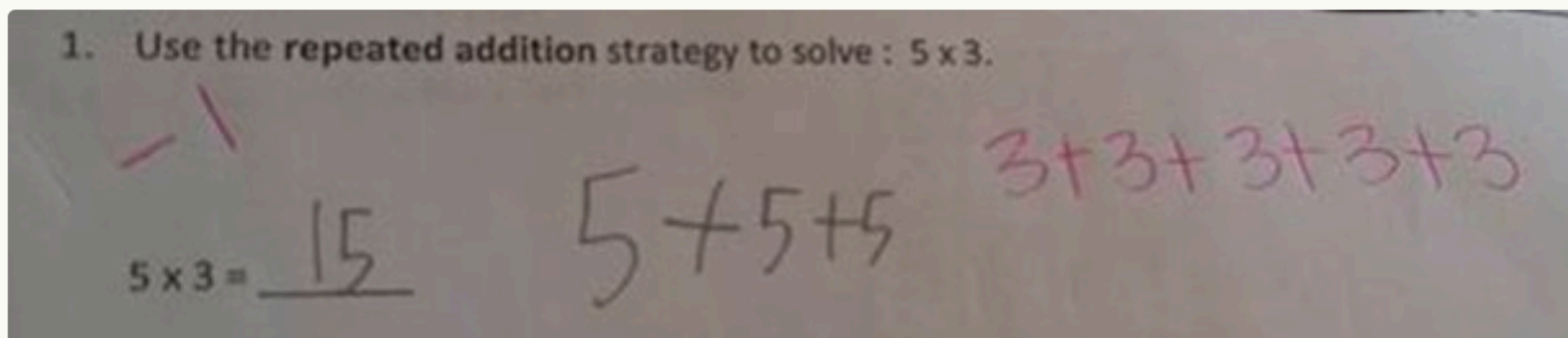
When you have one way to do it, hold up one finger. When you have two ways, hold up two fingers. And so on.

Some ways the room may have found: $8 \times 20 + 8 \times 4$, or $8 \times 25 - 8$, or 4×48 , then 2×96 . Every one of them gives 192.

Each of those has a name. By the end of today you will know all of them.

Math Broke the Internet

This photo of a third grader's paper went around the internet.



Talk with your shoulder partner. Is the child right? Is the teacher right?

In the equal groups model, 5×3 means 5 groups of 3, which is $3 + 3 + 3 + 3 + 3$. The child wrote 3 groups of 5, which is 3×5 . Both make 15, but they tell different stories. Was the mark fair?

Multiplying by a Power of 10

What happens when we multiply a number by 10?

With base ten blocks, every block becomes the next block up: each unit becomes a long, and each long becomes a flat.

With dots and boxes, every dot moves one box to the left. So $47 \times 10 = 470$.

Multiplying by 100 moves every dot two boxes to the left. So
 $3,406 \times 100 = 340,600$.

Six Rectangles Inside One

Use the area model to evaluate 246×73 .

	200	40	6	
70	14,000	2,800	420	73
3	600	120	18	
	246			

Expanded form: $246 = 200 + 40 + 6$
and $73 = 70 + 3$. That cuts the
rectangle into six pieces.

Find the six areas.

Add the six areas:

$$\begin{aligned} &14,000 + 2,800 + 420 \\ &\quad + 600 + 120 + 18 = 17,958 \end{aligned}$$

So $246 \times 73 = 17,958$.

The Areas Are the Partial Products

Procedure 2.4.13 Partial Products Algorithm for Multiplication

To multiply two whole numbers a and b using partial products, do the following.

1. Write a and b in expanded form.
2. Take each term from the expanded form of a and multiply it by each term from the expanded form of b . These are your partial products.
3. Add all the partial products.

Example 1 needed six partial products; two 2-digit numbers need four. What must a student already know for this to work?

Trade the Problem for an Easier One

Use additive compensation to evaluate 274×99 .

Which factor is close to a friendly number? 99 is one away from 100, and $274 \times 100 = 27,400$.

But that is 274 groups of 100, and each group is 1 too big. Compensate by taking 1 out of each of the 274 groups.

$$\begin{aligned} 274 \times 99 &= 274 \times 100 - 274 \times 1 \\ &= 27,400 - 274 \\ &= 27,126 \end{aligned}$$

When would you *add* to each group instead?

Halve One Factor, Double the Other

Which expression would the doubling and halving strategy work best on, 12×33 or 24×16 ? Why?

$$12 \times 33$$

$$6 \times 66$$

$$3 \times 132$$

3 is odd: stop, and finish another way.

$$396.$$

$$24 \times 16$$

$$48 \times 8$$

$$96 \times 4$$

$$192 \times 2$$

$$384 \times 1$$

16 halves all the way to 1: done. 384.

Both work. Doubling and halving works **best** when a factor can be halved all the way down.

Multiply One, Divide the Other

Group work. Multiplicative compensation: multiply one factor by a number and divide the other by the same number. Use it to evaluate your product.

Group member	Product
1	(a) 28×5
2	(b) 64×50
3	(c) 28×25
4	(d) 36×15

- Work your own product. Say which factor you made friendly and what you did to the other.
- Take turns. Show your group what you did, and ask questions to help your group members improve their explanations.

Answers: (a) 140 (b) 3,200 (c) 700 (d) 540

Breaking Apart Rests on Two Properties

Theorem 2.4.20 Associative Property of Multiplication of Whole Numbers

If a , b , and c are whole numbers, then

$$(a \times b) \times c = a \times (b \times c).$$

Theorem 2.4.22 Distributive Property of Multiplication Over Addition and Subtraction

If a , b , and c are whole numbers, then

$$a \times (b + c) = (a \times b) + (a \times c) \quad \text{and} \quad a \times (b - c) = (a \times b) - (a \times c).$$

Break the Big Factor Into Places

Use the breaking apart strategy to evaluate 7×134 .

Break 134 into its place values: $134 = 100 + 30 + 4$. Then apply Theorem 2.4.22.

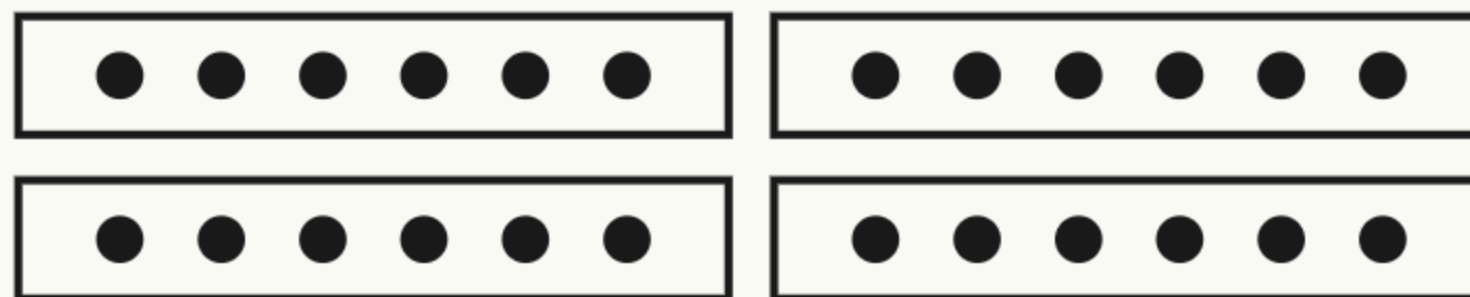
$$\begin{aligned} 7 \times 134 &= 7 \times (100 + 30 + 4) \\ &= (7 \times 100) + (7 \times 30) + (7 \times 4) \\ &= 700 + 210 + 28 \\ &= 938 \end{aligned}$$

In the warm-up, 8×24 became $8 \times 20 + 8 \times 4$, and it also became 4×48 .

Which property did each one use?

One Picture, Three Problems

Consider the following figure.



What do you see? **4 groups of 6.**

What **multiplication** problem could this figure be representing? $4 \times 6 = ?$

What **partitive division** problem could this figure be representing? $24 \div 4 = ?$
(24 shared into 4 groups; how many in each?)

What **quotative division** problem could this figure be representing? $24 \div 6 = ?$
(24 put into groups of 6; how many groups?)

Take Out a Group, Count It

Use the repeated subtraction strategy to evaluate $18 \div 3$.

Subtract a group of 3	Have left	Groups removed
$18 - 3$	15	1
$15 - 3$	12	2
$12 - 3$	9	3
$9 - 3$	6	4
$6 - 3$	3	5
$3 - 3$	0	6

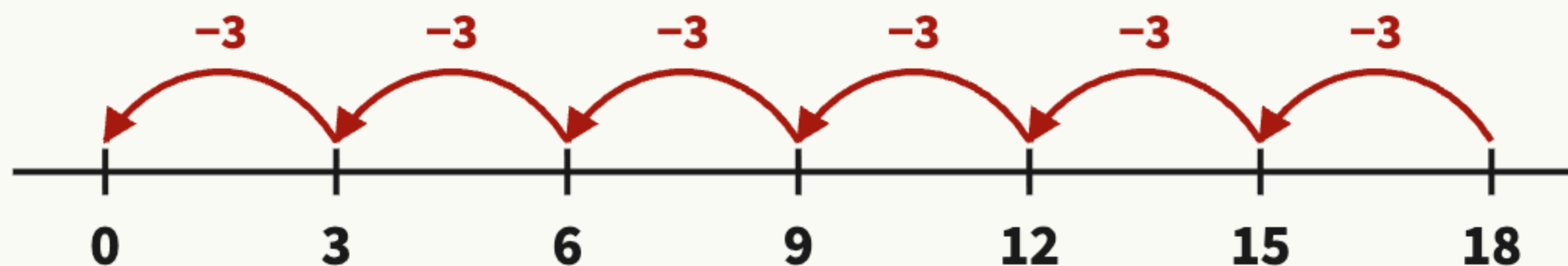
Six groups of 3 were removed, so $18 \div 3 = 6$.

Is repeated subtraction making use of the partitive or the quotative division model?

Quotative. We know the size of each group and count how many groups come out.

Skip Count Backward to Zero

Use skip counting to evaluate $18 \div 3$.



Start at 18 and count backward by 3's: 15, 12, 9, 6, 3, 0. Six backward steps of 3 equal one backward step of 18. So $18 \div 3 = 6$.

Is skip counting making use of the partitive or the quotative division model?

Quotative, the same as repeated subtraction. Each step backward removes one group of 3.

Remove Big Chunks of Groups

Group work. Use the partial quotients strategy to evaluate your group's problem.

Group	Problem
1	(a) $471 \div 6$
2	(b) $1,550 \div 23$
3	(c) $3,197 \div 8$
4	(d) $2,417 \div 37$

- As a group, write what 10 groups and 100 groups of the divisor would be.
- Agree on each chunk of groups to remove. Keep going until what is left is smaller than the divisor, then add up the groups you removed.

Answers: (a) 78 remainder 3 (b) 67 remainder 9 (c) 399 remainder 5 (d) 65 remainder 12

One Question to Leave With

Which strategy is the most efficient way to evaluate 44×25 ?

What is it about these numbers that makes your strategy efficient?

More than one answer is good. Multiplicative compensation gives $11 \times 100 = 1,100$, because $25 \times 4 = 100$. Breaking apart gives $40 \times 25 + 4 \times 25 = 1,000 + 100$, because 25s are easy to count by.